R Programming Practical Examination

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Part A: Basic Operations

A1 Create v as the sequence 5, 10 , 15, . . . , 50 using seq(). Print its length, sum and mean.

```
v <- seq(50)
length(v)
```

```
## [1] 50
```

```
sum(v)
```

```
## [1] 1275
```

```
mean(v)
```

```
## [1] 25.5
```

A2 Create w that repeats the vector c(1, 2, 3) four times, and z that repeats each element of c(1, 2, 3) four times. Print both.

```
w <- c(1, 2, 3)
z <- c(1, 2, 3)
```

A3 Compute and print: the remainder of 17 divided by 5, the integer division of 17 by 5, 2 to the power of 10, square root of 144, and pi rounded to 3 decimals.

```
remainder <- 17 %% 5
division <- 17 / 5
```

A4 Given a <- c(2, 4, 6 ,8) and b <- c(1 , 3, 5, 7), print a + b, a * b, the logical comparison a > b, and the dot product sum(a * b).

```
a <- c(2, 4, 6, 8)
b <- c(1, 3, 5, 7)
a + b
```

```
## [1]  3  7 11 15
```

```
a * b
```

```
## [1]  2 12 30 56
```

```
a > b
```

```
## [1] TRUE TRUE TRUE TRUE
```

```
sum(a * b)
```

```
## [1] 100
```

A5 Use paste() to produce “Student-1” “Student-2” “Student-3”. Then identify the datatype of 42, 42L, “42”, 41i, and TRUE. What are the outputs?

```r
paste("Student-1", "Student-2", "Student-3")
```

```
## [1] "Student-1 Student-2 Student-3"
```

```r
typeof(42)
```

```
## [1] "double"
```

```r
typeof(42L)
```

```
## [1] "integer"
```

```r
typeof("42")
```

```
## [1] "character"
```

```r
typeof(41i)
```

```
## [1] "complex"
```

```r
typeof(TRUE)
```

```
## [1] "logical"
```

Part B: Data Frames and Conditional Statements

B1 Create a data frame students with 8 rows and these columns (use stringsAsFactors = FALSE; sex is "M" or "F"):

```r
students <- data.frame(
id = 1:8,
name = c("Anna", "Ben", "Cara", "David", "Ella", "Frank", "Grace", "Hugo"),
sex = c("F", "M", "F", "M", "F", "M", "F", "M"),
section = c("A", "A", "B", "B", "C", "C", "A", "B"),
midterm = c(56, 72, 90, 70, 75, 54, 99, 88),
final = c(82, 95, 80, 75, 50, 58, 91, 74),
attendance = c(91, 92, 94, 71, 82, 80, 75, 90),
stringsAsFactors = FALSE
)
```

B2 Display the structure, dimensions, column names, and summary of students.

```r
str(students)
```

```
## 'data.frame':    8 obs. of  7 variables:
##  $ id        : int  1 2 3 4 5 6 7 8
##  $ name      : chr  "Anna" "Ben" "Cara" "David" ...
##  $ sex       : chr  "F" "M" "F" "M" ...
##  $ section   : chr  "A" "A" "B" "B" ...
##  $ midterm   : num  56 72 90 70 75 54 99 88
##  $ final     : num  82 95 80 75 50 58 91 74
##  $ attendance: num  91 92 94 71 82 80 75 90
```

```r
dim(students)
```

```
## [1] 8 7
```

```r
names(students)
```

```
## [1] "id"         "name"       "sex"        "section"    "midterm"
## [6] "final"      "attendance"
```

```r
summary(students)
```

```
##        id              name               sex                section     midterm
##  Min.   :1.00   Length   :8   Length   :8   Length   :8   Min.   :54.0
##  1st Qu.:2.75   N.unique :8   N.unique :2   N.unique :3   1st Qu.:66.5
##  Median :4.50   N.blank  :0   N.blank  :0   N.blank  :0   Median :73.5
##  Mean   :4.50   Min.nchar:3   Min.nchar:1   Min.nchar:1   Mean   :75.5
##  3rd Qu.:6.25   Max.nchar:5   Max.nchar:1   Max.nchar:1   3rd Qu.:88.5
##  Max.   :8.00                                           Max.   :99.0
##      final         attendance
##  Min.   :50.00   Min.   :71.00
##  1st Qu.:70.00   1st Qu.:78.75
##  Median :77.50   Median :86.00
##  Mean   :75.62   Mean   :84.38
##  3rd Qu.:84.25   3rd Qu.:91.25
##  Max.   :95.00   Max.   :94.00
```

B3 Add a column average = the mean of midterm and final for each student, rounded to 1 decimal.

```r
students$average <- round((students$midterm + students$final) / 2, 1)
students
```

```
##   id  name sex section midterm final attendance average
## 1  1  Anna   F       A      56    82         91    69.0
## 2  2   Ben   M       A      72    95         92    83.5
## 3  3  Cara   F       B      90    80         94    85.0
## 4  4 David   M       B      70    75         71    72.5
## 5  5  Ella   F       C      75    50         82    62.5
## 6  6 Frank   M       C      54    58         80    56.0
## 7  7 Grace   F       A      99    91         75    95.0
## 8  8  Hugo   M       B      88    74         90    81.0
```

B4 Add a logical column passed that is TRUE when average >= 70 and attendance >= 75.

```r
students$passed <- students$average >= 70 & students$attendance >= 75
students
```

```
##   id  name sex section midterm final attendance average passed
## 1  1  Anna   F       A      56    82         91    69.0  FALSE
## 2  2   Ben   M       A      72    95         92    83.5   TRUE
## 3  3  Cara   F       B      90    80         94    85.0   TRUE
## 4  4 David   M       B      70    75         71    72.5  FALSE
## 5  5  Ella   F       C      75    50         82    62.5  FALSE
## 6  6 Frank   M       C      54    58         80    56.0  FALSE
## 7  7 Grace   F       A      99    91         75    95.0   TRUE
## 8  8  Hugo   M       B      88    74         90    81.0   TRUE
```

B5 using ifelse(), add a column standing with these values based on average: - "Honors" if average >= 90 - "Good" if average >= 75 (and below 90) - "Needs Support" otherwise

```r
students$standing <- ifelse(students$average >= 90, "Honors", ifelse(students$average >= 75, "Good", "N
students
```

```
##   id  name sex section midterm final attendance average passed      standing
## 1  1  Anna   F       A      56    82         91    69.0  FALSE Needs Support
## 2  2   Ben   M       A      72    95         92    83.5   TRUE          Good
## 3  3  Cara   F       B      90    80         94    85.0   TRUE          Good
## 4  4 David   M       B      70    75         71    72.5  FALSE Needs Support
```

```
## 5  5  Ella   F       C      75    50         82    62.5  FALSE Needs Support
## 6  6 Frank   M       C      54    58         80    56.0  FALSE Needs Support
## 7  7 Grace   F       A      99    91         75    95.0   TRUE        Honors
## 8  8  Hugo   M       B      88    74         90    81.0   TRUE          Good
```

B6 Write a function classify(score) that uses and if else if else chain and returns "Distinction" for 85 or above, "Merit" for 70 to 84, "Pass" for 50 to 69, and "Fail" below 50. Test it with sapply(c(90, 73, 55, 40), classify) and print the result.

```
classify <- function(score) {
  if (score >= 85) {
    return("Distinction")
  } else if (score >= 70) {
    return("Merit")
  } else if (score >= 50) {
    return("Pass")
  } else {
    return("Fail")
  }
}

result <- sapply(c(90, 73, 55, 40), classify)
print(result)
```

```
## [1] "Distinction" "Merit"       "Pass"        "Fail"
```

**Part C: rbind() and cbind()

C1 Create a data frame new_students with two students . Its columns must match the original, including average, passed, and standing, computed the same way.

- id 9, Ines, F, section C, midterm 81, final 84, attendance 92
- id 10, Jamal, M, section A, midterm 70, final 68, attendance 80

Use rbind() to append them to students, saving the result as students_all.

```
new_students <- data.frame(id = c(9, 10), name = c("Ines", "Jamal"), sex = c("F", "M"), section = c("C"

new_students$average <- round((new_students$midterm + new_students$final) / 2, 1)

new_students$passed <- new_students$average >= 70 & new_students$attendance >= 75

new_students$standing <- ifelse(new_students$average >= 90, "Honors", ifelse(new_students$average >= 75

students_all <- rbind(students, new_students)
students_all
```

```
##    id  name sex section midterm final attendance average passed      standing
## 1   1  Anna   F       A      56    82         91    69.0  FALSE Needs Support
## 2   2   Ben   M       A      72    95         92    83.5   TRUE          Good
## 3   3  Cara   F       B      90    80         94    85.0   TRUE          Good
## 4   4 David   M       B      70    75         71    72.5  FALSE Needs Support
## 5   5  Ella   F       C      75    50         82    62.5  FALSE Needs Support
## 6   6 Frank   M       C      54    58         80    56.0  FALSE Needs Support
## 7   7 Grace   F       A      99    91         75    95.0   TRUE        Honors
## 8   8  Hugo   M       B      88    74         90    81.0   TRUE          Good
## 9   9  Ines   F       C      81    84         92    82.5   TRUE          Good
## 10 10 Jamal   M       A      70    68         80    69.0  FALSE Needs Support
```

C2 Create a numeric vector quiz <- c(8, 7, 10, 6, 7, 9, 10, 5, 8, 6). Use cbind() to attach it to student_all as a column named quiz. Confirm the result is still a data frame with class().

```
quiz <- c(8, 7, 10, 6, 7, 9, 10, 5, 8, 6)
```

C3 Create the matrix m <- cbind(midterm = students_all$midterm, $final = students_all$final) and use rbind() to append a final row containing the column means, labelled "mean".

C4 in a comment (2-3 sentences), explain what error rbind() produces if the data frames have different column names, and show the output if you safely catch it with tryCatch().

```
tryCatch(rbind(students, bad), error = function(e) {message("rbind failed: ", conditionMessage(e)); NULL
```

```
## rbind failed: object 'bad' not found
```

```
## NULL
```

Part D: Subsetting

Use students_all (with quiz) for all questions. Print every result. **D1** Extract the name column in three ways: $, [["name"]], and [, "name"].

D2 Extract rows 2 to 5 and only the columns name, midterm, final.

D3 Using logical subsetting, select students with attendance above 90

D4 Using subset(), select students of sex "F" in section "A" or "C" whose final is at least 75; return only name, section, and final.

D5 use which.max() and which.min() to report the name of the student with the highest and lowest average.

D6 Sort students_all by average in descending order, then print the top 3 rows.

Part E: Basic Factors

E1 Convert section and sex in students_all to factor(). Print levels() of each.

E2 Use table() to count students per sex and per section. - Try the two way table of sec by section. table(students_all$sex, $students_all$section). What is the output?

E3 Print as.integer() of the sex factor. In a comment, explain what the integers represent.

E4 Create a new factor sex_label from sex using factor() with levels = c("F", "M") and labels = c("Female", "Male"). Add it to students_all and print its levels().

E5 Print summary() of sex_label. Then, using logical subsetting and mean(), print the mean average for each level of sex_label.